国家级高新技术企业 HI-TECH

Opmaterial
海川新材


深圳市小梅沙海滨栈道，铺筑于2012年


杭州市标准堤塘西兴段，铺筑于2016年



江苏海川室外停车坪，铺筑于2016年


淮南市淮河大道，铺筑于2008年，摄于2016年

彩色路面解决方案
Solutions for Colored Pavement

深圳海川新材料科技股份有限公司

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Leader in Greenway Development
全国彩色沥青混凝土绿道开拓领先者

Chief Editor of National Standard for Colored Asphalt Concrete
国家标准《彩色沥青混凝土》主编单位

Guangdong Province Road Engineering Materials Engineering Technology Research and Development Center
广东省道路工程材料工程技术研究开发中心监制

我们的宗旨 Our aim has always been to

客户至上 服务第一

Customer is God Service First

以质量求生存 以科技求发展

The survival by quality The development by Technology



海川上海浦东技术与应用服务中心

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1、公 司 简 介

SELF INTRODUCTION


国家科学技术进步奖

证 书

为表彰国家科学技术进步奖获得者，特颁发此证书。

项目名称：环保型路面建造技术与工程应用

获 奖 者：深圳海川工程科技有限公司

奖励等级：二等

证书号：2013-J-223-2-01-005









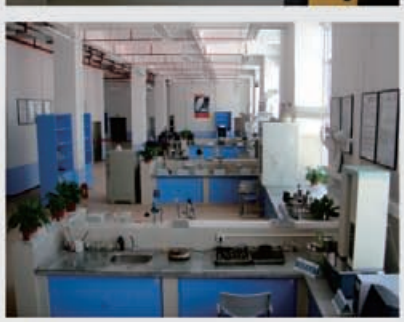




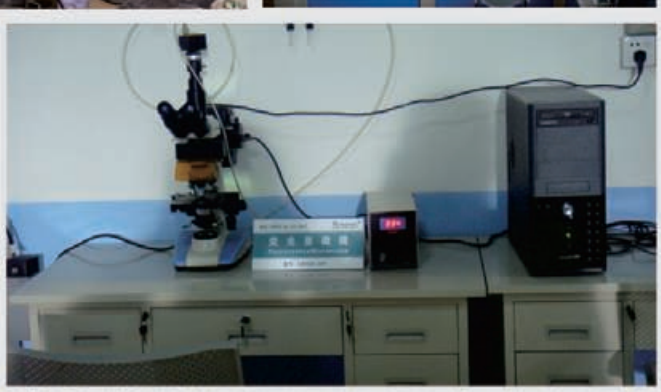












Opmaterial
海川新材

深圳海川新材料科技股份有限公司成立于2010年10月，是由深圳市海川实业股份有限公司控股的几个高科技全资子公司合并而成，新成立的深圳海川新材料科技股份有限公司共设立三个事业部、一个中央研发院、一个通过国家CNAL认证的材料检测中心，同时是SAC/TC 458/SC1秘书处的挂靠单位，其中3个业务部门前身分别是成立于上世纪90年代的深圳海川化工科技有限公司，海川色彩科技有限公司以及海川工程科技有限公司，这三家企业都是国家级高新技术企业，共拥有上海以及河源2个总占地为9万平方米的生产研发基地。

近年来，公司响应国家“节能减排”、“绿色低碳”的号召，在海川组建的“深圳市企业技术中心”、“博士后科研工作站”、“广东省道路工程新材料工程技术中心”等研发平台的基础上，通过承担“国家十一五科技支撑项目”以及建设部、交通部等科技项目迅速地提高了公司原创性技术开发、战略性技术研究的综合实力。多年来，通过探索和研发节能、低碳新产品，目前在节能、节水、节地、节材和环保上取得了系列的技术突破，并获得权威机构认证。产品目前也已经在全国得以广泛的推广和应用。

海川新材料科技股份有限公司注册资金为3333万，在全国设有30余个分公司，拥有遍及全国的销售和物流服务网络。目前共有员工200余人，70%为大专以上学历，其中高级工程师以及具有硕士以上学历的专业技术人员有18名，研发实力在同行之间名列前茅。

海纳百川，有容乃大，成为中国新型材料开发的领先者，致力于创造可持续发展的解决方案是我们的愿景和理念。挑战困难，以发展契机；完善自我，以创造明天是我们的精神。海川新材料科技股份有限公司立志为社会创造价值、为顾客创造价值、为股东创造价值、为员工创造价值，成为材料工业的创新型科技企业。

Shenzhen Opmaterials Co., Ltd was founded in October 2010, it is a consolidation of several subsidiary high-tech industrial companies of OceanPower Industrial Co., Ltd. Opmaterials Co., Ltd. sets up three career section, a central R & D Centre and a certified materials testing center by CNAL which is also the Secretariat affiliated units.of SAC / TC 4581 / SC1 The company has three business departments. which grew out of three state-level high-tech enterprise -- OceanPower Chemical Technology Co., Ltd.,OceanPower Color Technology Co. Ltd.and OceanPower Engineering Technology Co., Ltd. These three companies were founded in 1990s and has production research and development based in Heyuan and Shanghai, which cover an area of 90,000 square meters in total.

In recent years, the company has responded to the national calls of "energy saving" and "low carbon". By assuming the "National Eleventh Five Science and Technology Support Project "and the technology projects of Ministry of Construction and Ministry of Communications, Opmaterials has quickly improved its comprehensive strength in the exploration of original technology, and strategic technology research, and all projects are based on the R & D platform formed by Oceanpower, such as "Shenzhen Enterprise Technology Center" and "post-doctoral research station". Over the years, through the exploration and development of new energy-saving and low-carbon products, Opmaterials currently has made a series of technological breakthrough in the energy-saving, water-saving, land-saving, materials-saving and environmental protection, and has obtained certification form authorities. At present, those products have been widely promoted and applied around the whole country.

Opmaterials Co., Ltd has 33.33 million registered capitals, more than 30 branches around the country, and nationwide sales network and logistics services. Currently, the company has more than 200 employees, 70 percent have college degree or above, 18 among them are mid-level or senior engineer which in another words, they have master's or higher degrees. This makes our company to be one of the best R & D companies among peers.

Be tolerant to diversity, tolerance is a virtue, This Chinese old saying stems in the company's philosophy.Being the pioneer of Chinese new materials industry and striving for creating the sustainable solution is our development vision. Challenging the Difficulties for creating development opportunities; perfecting ourselves for a better future is our enterprise's spirit. Opmaterials determines to be an innovation enterprise in new material industries that creates value for employees, for customers, for shareholders, as well as for the whole society.



2、彩色沥青产品介绍
Colored Asphalt Products

彩色沥青混凝土路面作用 Functions of Colored Asphalt Concrete Pavement

主要应用于城市道路，如公园、小区、风景区、观光道路、广场、游乐场等，起到美化环境、改善道路环境、提升城市形象作用。用于公交专用道、交叉路口等，具有诱导交通、缓解驾驶疲劳、引导车流、增加交通安全的功能。彩色沥青混凝土路面色彩明亮、反射率高，能减少路面吸收热量、降低城市热岛效应。彩色透水沥青混凝土路面，排水降噪、消除积水，利于建设海绵城市。

It is mainly used in urban roads, such as parks, residential areas, scenic spots, tourist roads, squares, playgrounds and etc, It can beautify the environment, improve the road situation, and perfect the image of the city. It is also used in bus lanes, intersections, and so on for relieving driving fatigue, guiding vehicle stream and enhancing the traffic security. On the one hand, because of the bright color and high reflectivity, color pavement can reduce the heat of roads, and remit the urban heat island effect. On the other hand, Colored permeable asphalt concrete pavement also helps on the city drainage, noise reduction, elimination of stagnant water, and the construction of sponge cities.

彩色沥青混凝土路面优点 Advantages of Colored Asphalt Concrete Pavement

- ■ 引导车流，增加交通安全
  Leading traffic flow, avoiding traffic accidents
- ■ 排水降噪，建设海绵城市
  Decreasing the noise , draining the water, constructing a sponge city
- ■ 减缓疲劳，增加行车安全
  Reducing driving fatigue, increasing traffic security
- ■ 美化环境，提高城市品味
  Beautifying the environment, improving the image of the city
- ■ 增加反射，降低路面温度
  Increasing reflection , reducding temperature of road surface

彩色路面铺装分类 Colored pavement classification

按面层材料分 Surface Material Classification		特点 Feature
彩色沥青混凝土路面 Colored asphalt concrete pavement		柔性路面，受温度变化影响小，色彩持久稳定，价格适中，环保 Flexible pavement，Little affected by temperature changes，Lasting color stability，Affordable，Environmental protection
彩色聚合物路面 Colored polymer concrete pavement		颜色鲜艳，多色系，高摩擦系数，减少刹车距离，快速固化 Bright colors，Multi-color，High coefficient of friction，Reduce braking distance，Fast curing
彩色水泥混凝土路面 Colored cement concrete pavement		表层易磨耗，破坏后维修困难，价格低 Surface easy to wear，Difficult to repair after damage，Lower prices
彩色砌块式路面 Colored Block Pavement		砌块与下面层结合性差，容易发生砌块脱落现象，价格较低，色彩丰富 Block poor binding with layer below，Prone to the phenomenon of falling blocks,Lower prices,Rich color
彩色弹性塑胶路面 Color flexible plastic pavement		应用范围仅仅限于操场、跑道，施工繁琐，价格昂贵，不环保 Applications are only limited to the playground, runway, construction cumbersome, expensive and not environmentally friendly

路威彩色沥青结合料 RoadPower Color Asphalt Binder

彩色沥青结合料通常指的是浅色结合料，它是一种新型铺面材料，有着与常规沥青类似的性质，常把它称作彩色沥青。路威彩色沥青结合料是利用国外的先进技术，按照沥青混合料的国家标准研制而成，由于加入了高分子改性剂，在塑性方面有很大的提高，延性明显好于普通沥青。

Color binder usually refers to the combination of light-color material, it is a new paving material has similar properties to conventional asphalt, often call it the color asphalt.RoadPower Colored asphalt binder is the use of foreign advanced technology,in accordance with national standards developed from the asphalt, due to the addition of polymer modifier, has greatly improved in terms of plasticity, ductility significantly better than conventional asphalt .



彩色沥青结合料
color asphalt binder

路威彩色沥青着色剂 RoadPower Color Asphalt Colorant

依托海川新材建筑颜色设计与表示方法的雄厚实力，我们对彩色沥青混凝土路面铺装专用着色剂的色彩实行全方位的定位、包装和设计，使我们的产品在各个方面均能满足广大用户的要求。

Based on Opmaterials Co., Ltd, Architecture color design and representation prowess, we implement all-round of positioning, packaging and design asphalt pavement on the color of special colorant for paving of colored asphalt concrete pavement , so that our products can meet the demands of customers in all aspects.



彩色沥青着色剂
Colored asphalt coloring agent

路威彩色沥青结合料运输方式
RoadPower Colored Asphalt Transportation Modes

1、罐装运输（20-30吨/罐,可对接拌合楼）（推荐使用,方便快捷）
2、桶装运送（190kg/桶,需脱桶设备）

1.Heating filling transport (20-30 t / can be docked mixing floor) (recommended, convenient)
2.Bottled transport (190kg / barrel, the barrel off equipment required)



罐装运输
Heating filling transport

彩色沥青结合料技术指标 Color Asphalt Binder Specifications

指 标 Index			单 位 Unit	技术要求 Skills Requirement		
				OP-90	OP-70	OP-50
针入度 Penetration （25 ℃，100g，5s）			0.1mm	80～100	60～80	40～60
软化点 Softening Point（环球法） (R&B)	机动车道 Motor vehicle lane		℃	≥50	≥55	≥60
	其他 Others			≥45	≥46	≥49
延度 Ductility	机动车道 Motor vehicle lane (10 ℃)		cm	≥45	≥30	≥20
	其他 Others (15 ℃)			≥100		≥80
闪点 Flash point			℃	≥230	≥240	≥250
密度 Density (15 ℃)			$g \cdot cm^{-3}$	实测记录 Records Found		
60℃动力黏度	机动车道 Motor vehicle lane		Pa·s	≥160	≥180	≥200
	其他 Others			≥140	≥160	≥180
135 ℃运动黏度 135 ℃ Kinematic viscosity			Pa·s	≤3		
颜色等级（铁钴比色法）Color grades (Fe-Co colorimetric)			档 Grade	≤17		
TFOT（或RTFOT）后残留物 TFOT (or RTFOT) the remains	质量变化 Mass change		%	≤±1.5		
	残留针入度比 The residue penetration ratio		%	≥55		
	残留延度 Remain Ductility	机动车道(10 ℃) Motor vehicle lane(10 ℃)	cm	≥8	≥6	≥4
		其他(15 ℃) Others(15 ℃)		≥20	≥15	≥10
颜 色 Colour			—	无明显变化 No significant changes		



2、彩色沥青产品介绍
Colored Asphalt Products

着色剂技术指标 Colorants Technical Indicators

指　标 Index	单　位 Unit	技术要求 Skills Requirement
水溶物含量 Water soluble content	%	≤1.0
105°C挥发物 Volatile	%	≤1.5
着色力 Coloring	—	98~102
吸油量 Oil absorption	%	≤22
筛余物含量（0.045mm筛孔）	%	≤0.1
色光 Shade	—	近似微值 Approximate similarity

彩色沥青混凝土技术指标 Technical indicators of colored asphalt concrete

高温稳定性 High temperature stability	水稳定性 Water stability	色彩稳定性 Color stability

高温稳定性指标 High Temperature Stability

混合料类型 Mixture Type	马歇尔稳定度技术要求/kN Technical Requirements of Marshall Stability/kN		流值技术要求/mm Technical Requirements of Flow Value/mm	
	机动车道 Motor vehicle lanes	非机动车道 Non-motorized vehicles	机动车道 Motor vehicle lanes	非机动车道 Non-motorized vehicles
AC	≥8	≥3	1.5~4	2~5
SMA	≥6.0	≥5.5	—	2~5
OGFC	≥3.5		—	

混合料类型 Mixture Type	单位 Unit	动稳定度技术要求 Dynamic Stability Technical Requirements			
		机动车道 Motor vehicle lanes		其他 Others	
		彩色沥青混凝土 color asphalt concrete	改性彩色沥青混凝土 Modified color asphalt	彩色沥青混凝土 Color asphalt concrete	改性彩色沥青混凝土 Modified color asphalt
AC	次/mm	≥1000	≥2400	≥600	≥1000
SMA		≥1500	≥3000	—	
OGFC		—		≥600	≥1000

水稳定性 Water Stability

指标 Index	单位 Unit	技术要求 Skills Requirement	
浸水马歇尔残留稳定度 Immersion Marshall stability residue	%	机动车道 Motor vehicle lanes	其它 Others
		≥80	≥75
冻融劈裂残留强度比 Freeze – thaw splitting residual strength ratio		≥80	≥75

色彩稳定性 Color Stability

指标 Index	单位 Unit	技术要求 Skills Requirement
色彩 Color	分 Score	≥80

彩色路面铺装设计方案 Color Pavement Design

1、按施工领域分类 Classification According To The Construction Field

- ■ 人行车道 Pedestrian Lane
- ■ 轻型车道Light Transportation
- ■ 自行车道 Cycle Line
- ■ 停车场 Park Area
- ■ 市政绿道 Municipal Green Road
- ■ 景观道路 Landscape Road
- ■ 休闲广场 Entertainment Plaza
- ■ 工业园区 Industrial Park
- ■ 公交车专用车道 Bus Lanes
- ■ 高速公路服务区 Expressway service area

2、彩色层厚度 Color Layer Thickness

根据需要常用厚度在2cm~4cm Commonly, the thickness is 2cm~4cm.

3、彩色沥青混凝土路面典型结构 Typical Structure of Colored Asphalt Concrete Pavement



4、彩色层色彩 根据需要可以选择多种色彩

Variety of Colors Can Be Chosen According To Different Needs.





2、彩色沥青产品介绍
Colored Asphalt Products

配合比设计步骤 Mix Design Procedure

1 选定沥青及矿料
Asphalt and mineral aggregate selected

2 确定矿料配合比
Determine the mineral aggregate mix

3 确定沥青用量
Determine the asphalt content

4 级配设计完成后，混合料总质量2.5%的颜料替代部份矿粉进行确定最佳油石比的试验。
When the gradation design is complete,pigment that 2.5% by weight of the asphalt mixture substitute Partialslag ,then determine the optimum asphalt-aggregate ratio.

施工工艺 Construction Technology

- **施工要求** Construction Requirements

当气温低于10℃时，不宜进行彩色沥青混合料路面施工。如在0～10℃气温施工，必须采取确保施工质量的有效措施；在低于0℃及遇到大风的冬季不应施工，雨天不得铺筑彩色沥青混凝土路面。

When the temperature is below 10 ℃, the color is not suitable for asphalt pavement construction. As the temperature 0 ~ 10 ℃ construction must take effective measures to ensure the construction quality; below 0 ℃ or encountered strong winds winter construction should not construct, Must not pave the colored asphalt concrete pavement in rainy days.

- **施工温度** Construction Temperature

彩色沥青混合料施工温度 Color asphalt mixture temperature

项目 Items		温度/℃ Temperature / ℃
彩色沥青结合料加热温度 Colorless asphalt binder heating temperature		140～150
砂、石料加热温度 Sand, stone heating temperature		160～180
矿粉和着色剂 Slag and coloring agent		不加热 Unheated
施工气温 ≥10℃ Construction temperature ≥10 ℃	混合料出厂温度 Factory mix temperature	140～160
	送到工地时温度 To the site temperature	≥140
施工气温 0-10℃ Construction temperature 0-10 ℃	混合料出厂温度 Factory mix temperature	150～170
	送到工地时温度 To the site temperature	≥150

说明：由于不同着色剂耐温性存在差异，实际施工温度需结合着色剂进行调整。
Due to difference in temperature resistance between variable coloring agents, the actual construction temperature should be adjusted according to the coloring agent's features.

- **碾压要求** Rolling Requirements

压路机速度和遍数（参考） Road roller speed and the number of passes (reference)

压路机类型 Roller Type	压路机重量 Roller Weight	初压速度(m/min) First Press Speed (m/min)	中压速度(m/min) Final Pressure Speed (m/min)
钢(光筒)压路机 Steel (steel drum) Rollers	6～8t双轮 6～8t double wheel	25～35	40～55
振动压路机 Vibratory Roller	6～7t静压时 6～7t static pressure	25～35(静压) 25-35 (static)	30～35(静压) 30-35 (static)
大约遍数 About the number of passes		2	2～4 (至基本无轮迹) 2-4 (to almost no wheel tracks)

彩色防滑聚合物路面介绍 Introduction of Color Non-slip Polymer Pavement

彩色防滑聚合物路面在欧洲国家已被广泛使用，但在我们国家，这还是一个新鲜的事物。目前，在我国伴随着前所未有的交通流量的增长，交通堵塞的问题开始出现，特别是在整个城市中心地带。在欧洲，众多的交通管理的解决方案中，铺筑彩色防滑聚合物路面是其中主要的解决方法并被广泛应用于交通分流和交通减速提示。在坡道、弯道、隧道进出口、十字路口、人行横道、环岛、公交车专用道、公交车停靠站、高速公路收费口等交通事故多发地段，加铺彩色防滑路面可显著提高路面抗湿滑和防侧滑能力，可以起到很好的色彩警示作用和防滑作用。

Color Non-slip Polymer Pavement have been widely used in European countries, but in our country, it is still a new thing. At present, with the rapidly growth of traffic flow, traffic congestion problems begin to emerge, especially in the downtown area. In Europe, among the large number of traffic management solutions, the color of color Non-slip Polymer Pavement is the main method and is widely used in traffic diversion and traffic reduction tips.I In the high-accident districts, such as ramp, bend, import and export of tunnel, intersection, crosswalk, roundabout, bus lanes, bus stop station, highway toll entrance and so on, using color Non-slip Polymer Pavement can significantly increase the resistance of the pavement to reach the effect anti-slip, on the other hand, the bright color can also play the role of warning.

彩色防滑聚合物路面优点 Advantages of Color Non-slip Polymer Pavement

1 增加摩擦，减少刹车距
Increasing the friction, reducing the braking distance

2 耐磨时候，延长路面寿命
Enhancing the resistance, prolonging the life of roads

3 独特结构，降低行车噪音
Using Unique structure, reducing traffic noise

4 美化环境，提高城市品味
Beautifying the environment, improving the image of the city

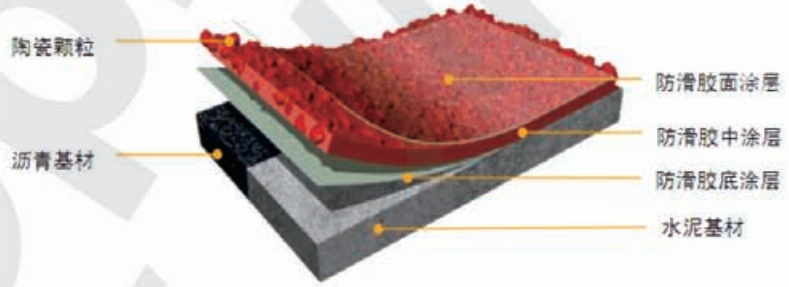
5 引导车流，增加交通安全
Leading traffic flow , increasing the traffic safety

6 增加反射，降低路面温度
Increasing reflection , reducing temperature of road surface

产品简介 Introduction

路威©彩色防滑聚合物路面是由彩色防滑聚合物路面胶A、B组份、耐磨骨料和抗紫外耐磨面漆组成，A组分和B组份按照一定比例拌合、摊铺后，均匀撒布彩色防滑集料；待固化后,扫除多余集料后在表面辊涂（或喷涂）一道抗紫外耐磨面漆，形成具有较大构造深度的彩色防滑面层。

Color Non-slip Polymer pavement is component by the glue A, B,abrasion and UV aggregate wear topcoat, A and B components in accordance with a certain percentage of mixing, after paving, spreading evenly color antiskid aggregates feed; after curing, remove excess aggregate surface after roller coating (or spray) a wear-resistant UV finish, non-slip surface to form a color layer having a larger construction depth.



彩色防滑聚合物路面胶 Technical Indicators of Colored Non-slip Polymer Pavement

彩色防滑聚合物路面胶是由海川自主研发的新型高分子材料，施工时A组份中加入B组份（固化剂），使体系产生快速的网状的交联反应，最终形成高强度高附着力的涂膜，从而达到在公路的特殊使用条件下的耐久使用性。

Glue for Color non-slip polymer pavement is made of Opmaterials independent research plastic and development of new polymer materials, construction component A was added B (curing agent), so that the system generates rapid mesh crosslinking reaction, forming high strength and adhesion coating film. So as to achieve the durability of highway under the particular conditions.



彩色防滑聚合路面胶
Glue for Color non-slip polymer pavement



3、彩色防滑聚合物路面
Color Non-slip Polymer Pavement

陶瓷颗粒 Ceramic Particles

常用颜色为红、绿、黄、蓝四大色系的道路面层，色彩丰富、持久，彩色防滑路面的抗滑值稳定地保持在70以上。路面系统所使用路面颜色由所选集料的颜色决定。

Common colors are red, green, yellow and blue colors of the four road surface, colorful, durable, non-slip surface of the skid color value remained stable at 70 or more. Color of pavement is determined by the selection of material color.



彩色防滑聚合物路面胶技术指标 Technical Indicators Glue Color Non-slip Surface

检验项目 Test Items	技术指标 Technical Specifications	
	非机动车道使用 The use of non-motorized vehicles	机动车道使用 Motor vehicle road use
基料在容器中的状态 Base status in the container	无结块、结皮现象、易于搅匀 No lumps, skinning, easy to stir	无结块、结皮现象、易于搅匀 No lumps, skinning, easy to stir
凝胶时间 min Gel time min	≥10	≥10
抗拉强度（25℃）MPA Tensile strength（25℃）MPa	≥6	≥7
断裂伸长率 % Elongation at break%	≥20	≥35

陶瓷颗粒技术指标 Ceramic Particles Technical Specifications

检验项目 Test Items	技术指标 Technical Specifications
莫氏硬度 Mohs hardness	≥6
骨料粒径，mm Aggregate size，mm	≤4

彩色防滑聚合物路面系统性能指标 Color Non-slip Surface System Performance

性能 Performance	实测值 Real Measurement values
构造深度/mm Texture Depth / mm	1.6~2.0
抗滑摆值 BPN Skid pendulum value BPN	75~85

彩色防滑聚合物路面厚度 Anti-skid Pavement Thickness Design

交通等级 Traffic Level	轻级 Light Level	中级 Intermediate	重级 Heavy
车流量Traffic	≤4500 PCU/日/车道	≤4500~6450 PCU/日/车道	≥6450 PCU/日/车道
路面厚度 Pavement Thickness	1~2mm	2~4mm	2~5mm

彩色防滑聚合物路面(施工工艺)

Color Non-slip Polymer Pavement (Construction Technique)

Opmaterial
海川新材





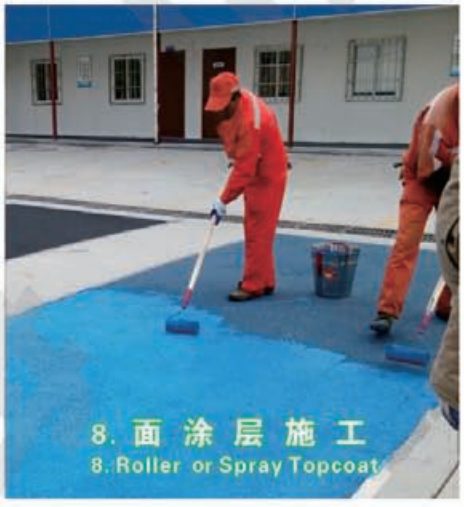
















4、彩色沥青修复剂
Colored Asphalt Repair Agent

彩色沥青修复剂 Colored Asphalt Repair Agent

彩色沥青在我国已经有十多年的发展历程，随着时间的推移，原来的彩色沥青混凝土路面已经老化、褪色，失去了原有的美观效果。我司自主研发的彩色沥青修复剂，由特种聚合物树脂和不同规格的集料组成，在老旧沥青混凝土路面上喷涂或辊涂一层，可恢复沥青混凝土路面的颜色，使其亮丽一新。其独特的化学结构，与彩色沥青形成良好的粘附效果以及三维网状结构，使彩色沥青修复剂有很好的耐磨、耐候性能。

The development history of colored asphalt is more than 10 years in China. With the passing of time, the original colored asphalt concrete pavement has been aged, faded, and has lost the original aesthetic effect. The self-developed color asphalt repair agent of Opmaterials is composed of special polymer resin and different specifications of the aggregate. By using a spraying or rolling in the old colored asphalt concrete pavement, colors can be restored and acquire a completely new look. Its unique chemical structure makes the color asphalt to have good adhesion and the three-dimensional network structure makes the color asphalt repair agent to have a good wear resistance and weather resistance.

用途 Application

应用于城市道路，如斑马线路面、整铺型路面、导向型路面、城市坡道、收费站专用车道、园林环境美化、自行车道、人行道，用于强化交通安全导引，体现城市人文精神。

Applied to urban road, such as a zebra crossing the road, the whole shop road, oriented road, city ramp, toll lanes, garden landscaping, bicycle lanes and sidewalks, guidance to strengthen traffic safety, reflect city humanistic spirit.

主要用于 Main Usage

1.用于彩色沥青混凝土路面的预养护或养护作业，可恢复原路面的亮丽颜色； 2.黑沥青混凝土路面的彩化；3.水泥混凝土路面的彩化。

1.For pre-curing or color pavement maintenance operation, restore the original color of bright color pavement. 2.Color of black asphalt pavement. 3.Color of the cement pavement.

厚度设计 Thickness Design

推荐0.5-1mm，可根据需要调整。

Recommended 0.5-1mm, can be adjusted as needed.

颜色 Color

可根据客户需要进行调整，常用色为红色、绿色、黄色、蓝色和灰色等。

can be adjusted according to customer needs, common color is red, green, yellow, blue and gray and so on.

技术分析 Technical Analysis

- 耐水隔水，延长路面寿命 Water-impermeable, extend pavement life
- 耐低温好，适合不同地区 Good low temperature resistance, suitable for different regions
- 施工方便，无需大型器械 Construction convenient, no large-scale equipment
- 粘结牢固，不脱落、剥离 Firmly bonded, not falling, peel

彩色沥青修复剂技术指标 Colored Asphalt Repair Agent Specifications

检验项目 Test Items	技术指标 Technical Specifications
基料在容器中的状态 Base status in the container	无结块、结皮现象、易于搅匀 No lumps, skinning, easy to stir
凝胶时间，min Gel time min	≥10
基料附着性（划圈法），级 Base adhesion (circle method)	≤4级 level
不粘胎干燥时间，h Sticky tires drying time, h	≤1
涂层低温抗裂性 Low temperature cracking resistance coating	无裂纹 No cracks
人工加速耐候性 Artificially accelerated weathering	不产生龟裂、剥落 No cracking, spalling

彩色沥青修复剂(施工工艺)

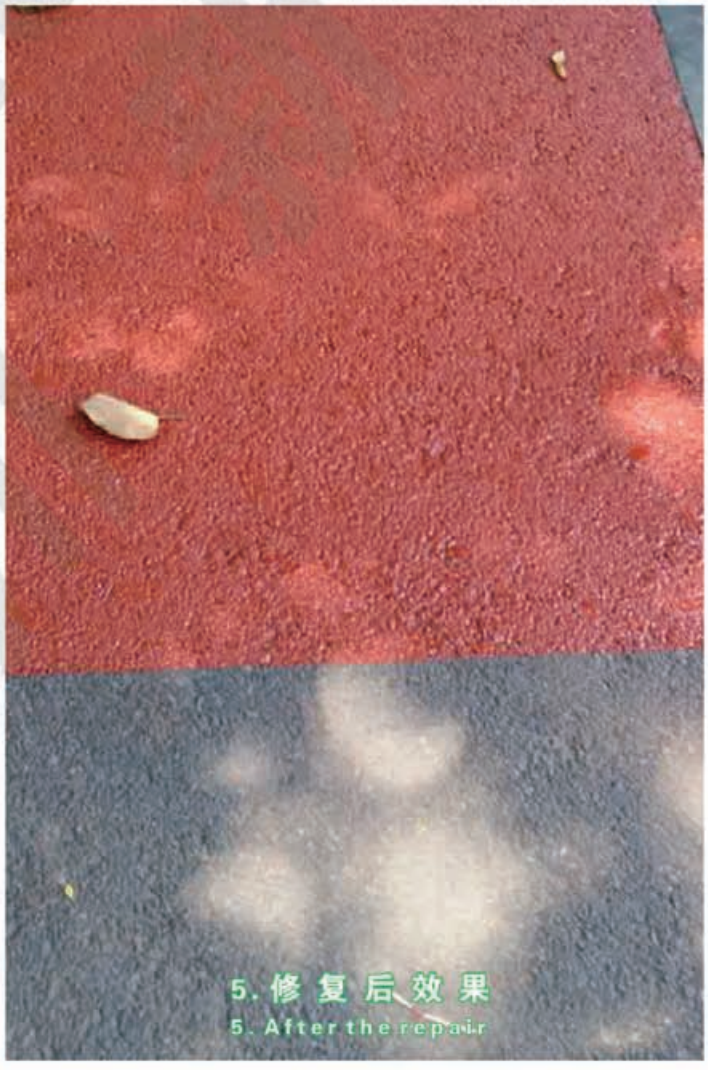
Colored Asphalt Repair Agent (Construction Technique)

Opmaterial
海川新材











5、部分工程案例

Part of The Engineering Case

Opmaterial 海川新材



山东潍坊福寿东街非机动车道

Weifang live on east non-motor vehicle lanes, Shandong Province



福建漳州城管大楼通道

Fujian zhangzhou city building channels



浙江杭州市钱塘江标准堤塘

Qiantang river road, Hangzhou City, Zhejiang Province



杭州湘湖旅游度假区湘堤彩色沥青项目

Xiang di colored asphalt Colored asphalt pavement, Xianghu lake tourist resort, Hangzhou province



山东潍坊胥山街彩色透水项目

Xushan street colored porous pavement, Weifang City, Shandong Province

5、部分工程案例

Part of The Engineering Case

Opmaterial 海川新材













5、部分工程案例

Part of The Engineering Case



2004年张家港市长安路

In 2014，Chang'an Road,Zhangjiagang City



张家港市长安路2016年现状，路面耐久性优良

In 2016，the status of Chang' an Road，Zhangjiagang City，Good durability of pavement



2008年淮南市淮河大道

In 2008，Huaihe Avenue，Huainan City



淮南市淮河大道2016年现状，路面耐久性优良

In 2016，the status of Huaihe Avenue，Huainan City，Good durability of pavement



浙江省杭州市钱江世纪城沿江景观带堤顶道路

Riverside Landscape Zone Embankment Top Road in Qianjiang Century City, Hangzhou, Zhejiang Province

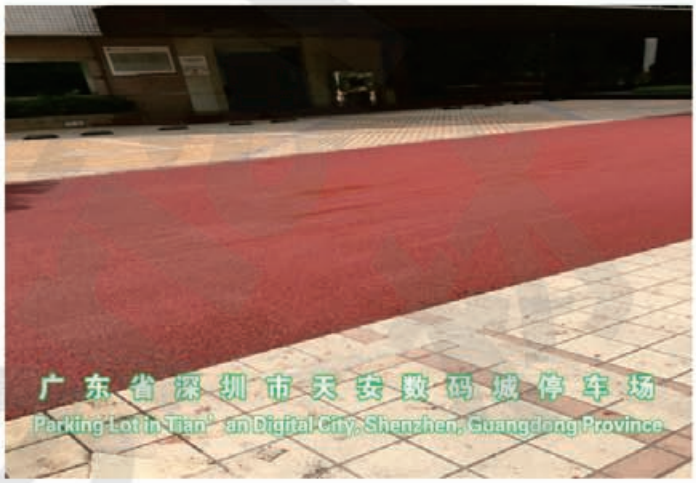


湖北省宜昌市BRT建设及东山大道改造工程三标段

BRT Construction and The Third Tender of Dongshan Avenue Renovation Project inYichang City, Hubei Province



杭州萧山瓜沥镇友谊路非机动车道
Non-motorized Vehicles Lane in Friendship Road, Guali Town, Xiaoshan, Hangzhou City



广东省深圳市天安数码城停车场
Parking Lot in Tian' an Digital City, Shenzhen, Guangdong Province



上海军工路公交枢纽站
Shanghai Jungong Road Bus Hub Station



该项目为青藏高原地区首条彩色沥青路面
青海西宁人民公园环湖路
People's Park Lake Road in Qinghai City, Xi' ning Province
The First Colored Asphalt Pavement in Tibetan Plateau



2005年福建厦门狐尾山步道
2005Foxtail Mountain Footpath in Xiamen City, Fujian Province



四川成都府城大道
Fucheng Road, Chengdu City, Sichuan Province



海口市滨海大道改造
Coastal Avenue Reconstruction in Haikou City



河南信阳南湾湖水库
South Bay Lake Reservoir, Xinyang City, Henan Province



上海市美兰湖景区步行道
Shanghai Meilan Lake Scenic Footpath



安徽淮南山南新区淮河大道非机动车道
Non-motorized Lane of Huaihe Road, Shannan South New District, Huainan City, Anhui Province



内蒙古包头民族东路
Nation East Avenue, Baotou City, Inner Mongolia Province



江西南昌青山路
Qingshan Road, Nanchang city, Jiangxi provice

5、部分工程案例

Part of The Engineering Case

Opmaterial 海川新材



深圳盐田区沙深路绿道

Green Path in Shashen Road, Yantian District, Shenzhen City



深圳盐田区环梅路绿道

Greenway of Huanmei Road, Yantian District, Shenzhen City



2014年深圳新彩通道，深圳市重点项目，SMA-13级配，海川新材提供全程技术支持

2014 Shenzhen Xincai Channel -- the Major project of Shenzhen City. SMA-13 Grading. The technical support was wholly provided by Opmaterials.



深圳市福田区福荣路绿道

Green Path in Furong Road, Futian District, Shenzhen City



深圳市福田区新洲绿道

Xinzhou Green Path, Futian District, Shenzhen City



深圳盐田区海景二路绿道

Green Path in Haijing 2nd Road, Yantian District, Shenzhen City

5、部分工程案例

Part of The Engineering Case

















Opmaterial
海川新材

















5、部分工程案例

Part of The Engineering Case

















5、部分工程案例

Part of The Engineering Case

















Opmaterial
海川新材


中山市中山五路绿道
Greenway of Zhongshan 5th Road, Zhongshan City


云浮市云城区沿江路绿道
Greenway of Riverside Road, Yunfu City


珠海斗门绿道
Doumen Greenway, Zhuhai City


珠海市省绿道II号线
The 2nd Lane of Province Greenway, Zhuhai City


珠海金鼎大学城项目
Jinding Univercity Project in Zhuhai City


茂名市茂南区滨海大道绿道
Greenway of Coastal Avenue, Maoming East District, Maoming City


河源公园路
Heyuan Park Road


佛山市顺德区陈村
Chen Village, Shunde District, Foshan City

6、客户服务

Part of The Engineering Case



过硬的服务团队

Excellent Service Team

海川是多年专业从事沥青材料研究的企业，技术服务团队有博士后、博士、硕士多人，大专以上学历人员100%，是国家级高新技术企业，技术团队经过多年的磨练，有着丰富的经验，能够给业主提供解决问题的可靠方案，能根据实际情况提供经济、实用、有特色的全面服务。

Oceanpower is an enterprise which has been specializing in research bituminous materials for many years. 100% of our staff has high degrees which cover from college degree to post-doctoral degree. Technical services team has a number of post-doctoral, doctoral, graduate researchers, all with college or higher educational background. As a national high-tech enterprise, the technical teams has a wealth of experience after many years of exercises, and can provide reliable, economic, practical, distinctive full-service solutions for the proprietors, according to the actual situation.

强大的技术支持

StrongTechnical Support

本公司引进欧美先进彩色沥青技术，在上海、深圳、河源设有专门的研发部门，与长安大学、重庆大学、长沙理工大学等高校有着密切的合作，公司一直走在产、学、研相结合的道路上，不断对彩色沥青产品进行研发改进，确保海川彩色沥青混凝土技术走在行业的前列。

By introducing the advanced color asphalt technology from Europe and the United States, with R & D departments in Shanghai, Shenzhen,and Heyuan, through the closely cooperation with universities, such as Chang'an University, Chongqing University, Changsha Technology University and etc., Opmaterials insists the developing road in the combination of production, learning and researching, strives for the continuous improvement and research of color asphalt products to ensure the color asphalt concrete technology be in the forefront of the industry.

Opmaterial
海川新材


DB44
广东省地方标准
DB44/T 420—2007
热拌热铺彩色路面胶结料、着色剂和集料
Binders pigments and aggregates used in color hot mixed asphalt pavement
2007-04-26 发布 2007-07-01 实施
广东省质量技术监督局 发布


UDC
中华人民共和国行业标准
CJJ
CJJ/T 218-2014
城市道路彩色沥青混凝土路面技术规程
Technical specifications for colored asphalt concrete pavement of urban road
2014-09-29 发布 2015-05-01 实施
中华人民共和国住房和城乡建设部 发布


GB
中华人民共和国国家标准
GB/T 32984-2016
彩色沥青混凝土
Color asphalt concrete
2016-08-29 发布 2017-07-01 实施
中华人民共和国国家质量监督检验检疫总局
中国国家标准化管理委员会 发布

可靠的质量保证
Reliable Quality Assurance

海川彩色沥青项目已经有十几年的实际工程经验，截止目前，海川彩色沥青施工面积已经超过一千万平方米，工程项目遍布华南、华东、华中、东北、西北、西南等全国近30个省市自治区。海川从铺装方案设计、彩色沥青胶结料加工、彩色沥青胶结料配送到工程施工质量控制有着一套完整的流程，确保彩色沥青路面铺装质量。

Oceanpower color asphalt project has more than 10 years of practical experience, until now, Haichuan color asphalt construction area has more than ten million square meters, the project over the South China, East China, Central China, Northeast, Northwest, Southwest across nearly 30 provinces municipalities and autonomous regions. Haichuan has a complete set of processes from the pavement design, color asphalt binder processing and distribution to the construction project quality control,to ensure the quality of the color asphalt paving.

完善的标准体系
Perfect Standard System

参编主编各项彩色沥青混凝土标准，2006年开始主编国内第一部地方性彩色沥青标准DB 44/T 420-2007《热拌热铺彩色路面胶结料、着色剂和集料》，填补了行业空白，为彩色沥青在广东乃至全国的推广奠定了基础；海川是住建部发布的《城市道路彩色沥青混凝土路面技术规程》CJJ/T 218-2014第一参编单位；主编完成国家标准《彩色沥青混凝土》GB/T 32984-2016。同时，公司为全国沥青混凝土标准化技术委员会秘书处承担单位，技术委员会同行业顶尖专家云集，对各项标准进行严格把关，使我国彩色沥青标准逐渐与国际标准接轨，也使得公司本身产品质量不断升级，具备国际竞争实力。

Oceanpower directed or participated in multiple color asphalt concrete standard editions. In 2006, Oceanpower hosted the first national local color asphalt standard DB 44 / T 420-2007 "Hot shop hot mix pavement binder color, coloring agents and aggregates" which filled gaps in the industry. This provided a foundation of the development of colored asphalt in Guangdong province, even in the whole country; Oceanpower was assigned by the Ministry of Housing to be the one of organizations who can firstly participate in the edition of "Urban road technical Specification color asphalt concrete pavement" CJJ / T 218-2014and "color asphalt concrete" GB/T 32984-2016. Meanwhile, the company is the commitment unit of the National he National Asphalt Concrete Standardization Technical Committee Secretariat. The Technical Committee and the industry' s top experts gathered together to strict check and adjust asphalt standards, so that our standard of colored asphalt can gradually meet the international standards. This also makes the company's own products quality continuously upgrading. As a result, our products have international competitiveness.

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http://www.opxincai.com

海川彩色沥青发展历程:

Oceanpower colored asphalt development course

2004年至今，深圳海川新材料科技股份有限公司彩色沥青混凝土路面铺装总面积约10,000,000平方米，项目涉及机动车道（含公交专用车道和高速公路服务区）、非机动车道、人行道、园林景观道路等。其中，我司于2004年在张家港市长安路铺筑的彩色沥青混凝土路面，经过十余年考验，至今路面使用效果优良，是我司彩色沥青混凝土路面应用的典型项目之一。

Since 2004, Shenzhen Opmaterials Co., Ltd. has paved colored asphalt condete pavement for about 10,000,000 square meters. The projects includes motor ways (including bus lanes and expressway service area), non-motor ways, sidewalks, landscape roads, etc.A colored asphalt concrete pavement, constructed by OceanPower at the location of Chang'an road, Zhangjiagang, 2004, has been used for nearly ten years. By now, the service performance is still superior. This project is one of the most representative project in the field of using colored asphalt concrete pavement.

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National level high-tech enterprises

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ISO 14001 SGS
SWISS CERTIFICATION Accreditation No.024